Project brief: Design a PCBs + order prototype + test prototype

PCB#2 is a **carrier board** for a Raspberry PI Compute Module 5. It shall connect to a NVMe drive, two Raspberry Global Shutter Camera, two MOSFET 24V switches (to control a fan and heater) and a I2C dew sensor. The PCB shall be powered from a 24V input, and therefore needs to have two Texas Instrument PowerPad buck converters (see below).

No coding is included in this project. However, the coder must upload (to each board) very simple **test software** (provided by me in the appendix of this document) to prove in a video that the prototype is working correctly.

Please note that you may need to have the following hardware (if you don't have it already), in addition to oscilloscope and other professional equipment. Take that into consideration when you make your offer!

	Price in India
Raspberry PI CM5 (any version)	₹9,770 incl shipping Silverline Electronics, Element14 India
Raspberry PI official carrier board for CM5 (for programming)	₹2,336 Silverline Electronics, IndiaMART
M-key NVMe 2280 (any capacity, any Gen, but compatible with Raspberry PI, when it comes to booting as well (check a Raspberry compatibility list before buying). (NOTE: NVMe, NOT a SATA-based M.2 or a B-key)	₹2,739 Rees52, Amazon India
A non-PWM computer fan (preferably 60x60x25 but any size would work). Must have a standard 2-pin, 3-pin or 4-pin header.	₹79 - ₹199 Amazon India (SP Electron)

This is the brief. If you are interested, please check the full specification.

/Glenn Adelsten

Introduction & background

I am developing a system for sport analysis, consisting of two camera stands. Each stand will consist of a pole, a Raspberry Compute Module 5 (with two camera modules connected), and an ESP32 which acts as a combined watchdog (for the Raspberry PI CM5) combined with a speech generator for an attached speaker to call the score.
Therefore, I need the carrier board for said RPI CM5 to be designed. This PCB will henceforth be called PCB#2. The ESP32 board is called PCB#1 and is separate from this project.

The camera pole is high, therefore 24 volts is used for the two power leads. Upon arrival to the Raspberry PI board, using buck converters, 24V gets converted to the voltages the top system requires.

The ESP32 (PCB#1) and the Raspberry (PCB#2) will communicate over wifi (ad hoc mode) so that the "watchdog" can tell when the Raspberry has become unresponsive, and then reboot it by toggling its 24V and GND wires briefly.

Please add test points as recommended on both boards, for the 3.3V and 5.15V rails.

The PCB#2 needs to follow the size constraints of the hand drawing (see separate document).

Milestone/deliverable #1: Schematic and BOM for **PCB#2.**

Milestone/deliverable #2: 2D or 3D renders of the board, and a screenshot of the layout.

Milestone/deliverable #3: Gerber files, drill files and CPL files for **PCB#2.**

Milestone/deliverable #4: Prototype
Assist me in placing an order for working **prototypes** of PCB#2, **at my expense**, for delivery to you.
When you receive the prototypes, you will install the custom demo code that I provide in the appendix, to prove that hardware works.
Deliverable #4: Evidence video showing the whole system in view the whole time.
Manuscript: The power will be connected. The test software will then automatically start, and show how fan, LED outputs, audio and the e-ink display all works correctly. Put the video camera on a stand and let the test progress for at least 10 minutes.

Milestone/deliverable #5: Delivery of the prototype PCB#2 arrives to me.

Baseboard for Raspberry Pi CM5 (PCB#2)

This is a baseboard for Raspberry PI Compute Module 5 (CM5), having the following functions:

A) It shall hold the **CM5**, using the required 100-pin mezzanine high density connectors.
Use recommendations and references from Raspberry PI. Do not forget to connect GPIO_VREF and CM5_3.3V.

B) It shall hold an **NVMe** (2280) on the other side of the PCB.
Use a standard/high M.key connector (4.2 mm height).
Add **vias for cooling** the NVMe controller of the NVMe drive;
Use a final diameter of 0.3 mm (approx. 12 mil).
Space them in a grid at a distance of 0.8 mm to 1.2 mm (center-to-center) to avoid weakening the board or causing assembly issues like solder wicking.
To allow overclocking, it is important that the two wires in each differential pair is length-matched within 0.1 mm to prevent timing issues. The impedance must be matched precisely (90 Ohms of differential impedance).
Traces must be kept as short as possible.

C) Two ordinary "**mini FPC connectors**" (i.e. 22-pin, 0.5mm pitch MIPI CSI-2 connectors, as used by Raspberry PI 5), commonly labeled by Raspberry as CAM/DISP0 and CAM/DISP1.
Camera MIPI CSI-2 traces must be differential pairs (100 Ohms) and length-matched within 0.5 mm to ensure high-speed video stability.

D) Use Kyocera 2-pin **terminal block** (009276002021106) for the **24V power input**. Place protection component(s) here.

E) **GPIO**

- Two 3-pin DuPont connectors for external **dual-color** Chanzon 5 mm **LEDs**, red and green. Used for status indication for troubleshooting purposes. Connected via MOSFET.
  CHANZON, 5 mm
  Lens: Clear Lens
  Emitting Color: Red and Green Bicolor
  Common anode
  Viewing Angle: 30 Degree (Clear Lens)
  Forward Voltage / Current: R:1.8V G:3V-3.2V | 20mA (each color)

- **Password reset "button"** – N/C
  SMD magnetic switch such as Coto CT10-1540-G1 (positioned as shown on hand drawing)
  Suggested GPIO: 7 (clean, general purpose, default internal pull-up). Note: Must be 3.3V tolerant!

- **I2C dew point sensor** located near PCB edge as shown, as far away as possible from other components that causes heat, on a "heat island" with physical thermal relief slots (air gaps) around it, preventing heat to be transferred to the sensor via copper.
  Suggested: **SHT40**.
  Used GPIO: 2 (SDA), 3 (SCL)

- **PWM heater control**
  MOSFET to allow PWM-switched 24V power to two 5 watt 25JR20E heater resistors positioned at the outer edges of the PCB (see sketch). The 29 mm wire (on each side) of this model, will allow them to be placed to hover away from the PCB, closer to the camera lenses.
  Suggested GPIO: 12 (PWM0)

- **PWM fan switch** to allow Raspberry software to control a 24V fan.
  The built in fan control is circumvented and other GPIOs used, because our fan is not a “true PWM fan” (even though it will be driven from a PWM/regulated voltage).
  Pin 1 (GND) connected to DRAIN of MOSFET.
  Pin 2 (V+) connected to 24V.
  Pin 3 (tachometer) connected to a CM5 GPIO (pin 4, 5 or 6). No extra components, the internal pull up will be used.
  Pin 4 (PWM) still connected to the CM5 fan pwm pin (this will not be used for 2 pin fans, even though pwm technology will still be used).
  Connect SOURCE of MOSFET to common ground.
  GPIO pin to GATE of MOSFET through a 220 ohm resistor.
  Connect a 10 kOhm resistor from GATE to ground.
  Use **4-pin fan socket** (Molex **KK 254** part 47053-1000, or similar), even though the fan will probably be 2-pin (the 4-pin socket is backwards compatible).
  Suggested GPIO: 13 (PWM1)
  Connect flyback diode across fan terminals (cathode/stripe to 24V, anode to MOSFET DRAIN (this could protect against voltage spikes).

- **Dual N-channel MOSFET**
  Instead of using two MOSFETs, it might be better to use a dual version.

F) **Power conversion**

- Convert **24V to 3.3V 3A, dedicated for for the NVMe**.
  Use **TPS54334** because it is available in the same 8-pin SO PowerPad (DDA) package.
- **Convert 24V to 5.15V 5A for the Raspberry CM5.**
  Buck converter to adjust 24V input to drive the Raspberry CM5.
  (Internal note: PCB area required for this functionality: 30x30 or 15x60 mm)
  Use buck converter chip Texas Instruments **TPS54560B-Q1** (8-pin HSOIC). Set it to 5.15V. Locate it closest to the wall.
  Use 4-layer PCB, standard FR4, using standard 1 oz (35uM) copper and green solder mask.
  Use internal power (5V) and ground planes.
  Use via stitching to connect buck converter chip output to internal plane using multiple vias (at least 4-6).
  The thermal pad must be soldered to a large copper ground plane with at least 9-16 thermal vias (0.3 mm diameter) stitching to the bottom layer.
  Overspecify the inductor to 7 or 8 ampere.
  Use 4 layer PCB for best cooling.
  Heat distance - ensure there is at least 10 mm lateral distance between this chip and the hottest part of the NVMe drive (i.e. the NVMe controller chip) which is located on the other side of the PCB.
  Do a copper pour on the bottom layer specifically for the NVMe controller area, to act as a heat spreader. Even if there are no components there.
  Consider adding a “thermal chimney” (a dense grid of open-mask vias) directly under the CM5’s PMIC and SoC areas, stitching to the internal ground plane to help sink heat away from the mezzanine connectors.
  Power sequencing - Ensure the “soft start” capacitor (Css) is sized to provide a smooth ramp-up to 5.15V to prevent inrush current from triggering the CM5’s internal brownout protection.

G) **PCB**

There shall be two different BOM versions of PCB#2. They use the same layout, but the chip versions are different:

BOM A - optimal component values for high heat installations
TPS54560BQDDAQ1 (automotive version, load dump version good for spikes, -40 to 125 Celsius ambient, featuring PI coating against moisture and environmental stress)
243 kOhm resistor from RT/CLK to GND
10 uH inductor (saturation current 7A, DCR<20mOhm, i.e. Coilcraft XAL7070-103MEC.
200 uF (two 100 uF low-ESR ceramic (X7R) in parallel, to handle ripple), 10V X7R 1210
Input cap: 10 uF ceramic near chip, 100 uF electrolytic (bulk).
Catch diode: 60V, 5A (B560C-13-F).
Voltage configuration resistors: Upper 54.9 kOhm, lower 10.2 kOhm (0603).
Compensation network: 15 kOhm, 3.3 nF, 150 pF (0603)
Bootstrap: 0.1 uF ceramic (between BOOT and SW)
BOM B – for mild weather (recommended for where ambient temperature never increases above 40-50 degrees Celsius)
Use TPS54560BDDA version instead.
Vishay **IHLP4040DZER100M11**
Standard X7R ceramic

Conformal coating recommended, preferably silicon-modified acrylic (for example MG Chemicals 422C). Paint the 5.15V voltage configuration resistors and power section with a brush. Avoid the CM5 connectors!
Use “No-Clean” flux to prevent trapped ionic contaminants from causing dendritic growth under the coating.

H) **Protection**

Fan induction protection
Fast flyback diode, anode connected to -, cathode connected to +24V, located close to the fan.

Voltage oscillation protection (due to long cable and the fact that the incoming 24V power is switched using a MOSFET)
Implement a snubber circuit: 100 ohm resistor + 0.uF capacitor, in series, placed in parallel with the MOSFETs
Use a **second snubber** too – TVS diode (P6KE33A), also connected between + and -.
Decoupling capacitor 100μF, connected between + and – near CM5 power.

Input protection
For **surge suppression**, use a TVS diode (such as SMBJ28A) near the 24V connection points, and also a **reverse polarity protection** diode (such as PMEG4010) and a **fuse**.

Bulk capacitor (low-ESR 470 uF or 1000uF polymer) at the CM5 input (to absorb camera startup spikes).

Potentially useful links:

Raspberry CM5 datasheet: https://pip.raspberrypi.com/documents/RP-008180-DS-cm5-datasheet.pdf
2.7” Waveshare EINK raw display https://www.amazon.com/dp/B07RM2C833

Summary of GPIOs for PCB#2 (CM5)

Component	GPIO (BCM)	Function
PWM Fan	13	PWM1
PWM Heater	12	PWM0
Magnet Sensor	7	Input (Pull-up)
SHT40 Sensor	2 (SDA), 3 (SCL)	I2C-1
Red LED	26	Output
Green LED	20	Output

Python test code for PCB#2

Put the following Python script onto the NVMe drive and make it autostart.

The SHT4x library and Blinka must be installed via pip for this test script to run.

```
import time

import board

import busio

import adafruit_sht4x

from gpiozero import PWMOutputDevice, Button, LED


# --- CM5 PIN MAPPING (BCM) ---

FAN_PIN = 13            # PWM1

FAN_TACHO_PIN = 16      # Dedicated CM5 Tacho Pin

HEAT_RESISTOR_PIN = 12  # PWM0

RED_LED = LED(26)

GREEN_LED = LED(20)

MAGNET_SENSOR = Button(7, pull_up=True)


# --- SETUP ---

i2c = busio.I2C(board.SCL, board.SDA)

sensor = adafruit_sht4x.SHT4x(i2c)

fan_pwm = PWMOutputDevice(FAN_PIN, frequency=25000)

heat_pwm = PWMOutputDevice(HEAT_RESISTOR_PIN, frequency=100)

tacho_input = Button(16, pull_up=True) # Software pull-up enabled


def flash_leds():

    """Alternating flash for magnet sensor trigger."""

    for _ in range(5):

        RED_LED.on(); GREEN_LED.off(); time.sleep(0.1)

        RED_LED.off(); GREEN_LED.on(); time.sleep(0.1)

    GREEN_LED.off()


MAGNET_SENSOR.when_pressed = flash_leds


def run_test():

    print("CM5 Prototype Test Running...")

    while True:

        temp, humidity = sensor.measurements

        print(f"Temp: {temp:.2f}C, Humidity: {humidity:.2f}%")
```

```
        # Test Fan and Heater cycle
        print("Testing Fan...")
        fan_pwm.value = 1.0
        time.sleep(3)
        fan_pwm.value = 0

        print("Testing Heater...")
        heat_pwm.value = 0.5
        time.sleep(3)
        heat_pwm.value = 0

        time.sleep(10)

if __name__ == "__main__":
    run_test()
```